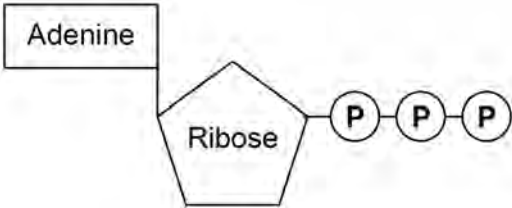


Question	Marking Guidance	Mark	Comments
02.1	1. Adenine, ribose and 3 x phosphate correctly labelled; 2. Bonds (between ribose and adenine and between ribose and phosphates) in correct positions; 	2 (2 x AO1)	1. Accept P in a circle/P_i/PO_4^{-3} for phosphate. Reject phosphorus 1. Ignore pentose, reject deoxyribose 1. Ignore labelling of bonds 1. Ignore shapes of adenine and phosphates
Question	Marking Guidance	Mark	Comments
02.2	1. Hydrolysis; 2. ATP hydrolase; 3. Phosphorylate other substances to make them more reactive/change their shape OR To form DNA/RNA/phospholipids;	3 (3 x AO1)	1. Reject hydrolysis of hydrogen bond 1. Ignore other named bonds 2. Accept ATPase 3. Accept description of phosphorylation 3. Accept any named biological molecule containing phosphate e.g. triose phosphate 3. Ignore to form ATP
Question	Marking Guidance	Mark	Comments
02.3	1. Sodium ions actively transported out (of cells); 2. Maintains/generates a concentration/diffusion gradient (for sodium ions); 3. <u>Co-transport</u> (with sodium ions);	3 (3 x AO1)	1, 2 and 3. Accept Na^+ for sodium ions 1. Accept 'pumped using ATP' for active transport 1,2 and 3. Reject lack of 'ions' once 3. Accept 'co-transporter' for 'co-transport'